Xinyuan Song

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Education

Peking University, BS in Physics (double major in Mathematics). Sept. 2014 – Jul. 2018
National University of Singapore (NUS), phD in Statistics. Sept. 2018 – May. 2023
University of Texas as Arlington, PhD in Computer Science. Sept. 2023 - Sept. 2024
Emory University, PhD in Computer Science (Adviser: Liang Zhao). Sept. 2024 –

Industry Experience

- **Alibaba Group** Jun. 2020 – Sept. 2021
  Machine Learning Engineer (full-time)

Academic Experience

- **Bayesian Statistics, PhD student** Sept. 2020 – Sept. 2023
  Instructor: Cheng Li, NUS
- **Manifold Learning, PhD student** Sept. 2020 – Sept. 2023
  Instructor: Zhigang Yao, NUS
- **LLM for medical, PhD student** Sept. 2023 – Sept. 2024
  Instructor: Yingying Zhu, University of Texas at Arlington
- **Multi-agent system, PhD student** Sept. 2024 –
  Instructor: Liang Zhao, Emory University

Papers (published)

1. **Xinyuan Song**, Ziyi Ni, Fred Yang, Bo Zhang, Yiming Yan, Yijin Wang, Jane Zhang. **P-QuASAR: A Unified Probabilistic Framework for Holistic Patent Quality Assessment and Refinement** *the Annual Meeting of the Association for Computational Linguistics (Findings)*, 2026.
2. Jiacheng Shi, Hongfei Du, **Xinyuan Song**, Y. Alicia Hong, Yanfu Zhang, Ashley Gao. **EffectCodec: Emotion-Preserving Neural Speech Codec for Expressive Speech Modeling** *the Annual Meeting of the Association for Computational Linguistics (Findings)*, 2026.
3. Wenfang Sun*, **Xinyuan Song***, Pengxiang Li, Lu Yin, Yefeng Zheng, Shiwei Liu **The Curse of Depth in Large Language Models** *NeurIPS*, 2025. (*equal contribution)
4. Hongyang He*, **Xinyuan Song***, Yangfan He, Zeyu Zhang, Yanshu Li*, Haochen You, Lifan Sun, Wenqiao Zhang, **TRiCo: Triadic Game-Theoretic Co-Training for Robust Semi-Supervised Learning** *NeurIPS*, 2025. (*equal contribution)
5. **Xinyuan Song**, Ziye Ma, **Matrix Sensing with Kernel Optimal Loss: Robustness and Optimization Landscape** *CPAL*, **Oral**, 2026.
6. **Xinyuan Song**, Yangfan He, Sida Li, Jianhui Wang, Hongyang He, Xinhang Yuan, Ruoyu Wang, Jiaqi Chen, Keqin Li, Kuan Lu, Menghao Huo, Ziqian Bi, Binxu Li, Pei Liu, **Efficient Temporal Consistency in Diffusion-Based Video Editing with Adaptor Modules: A Theoretical Framework** *CPAL*, 2026.
7. **Xinyuan Song**, Keyu Wang, PengXiang Li, Lu Yin, Shiwei Liu, **Demystifying the Roles of LLM Layers in Retrieval, Knowledge, and Reasoning** *ICASSP*, 2026.

8. Hongyang He*, Yan Zhong*, **Xinyuan Song***, Daizong Liu, Victor Sanchez, **Token-Aware Representation Augmentation for Fine-Grained Semi-Supervised Learning** *CPAL*, 2026. (*equal contribution)

9. Bo Gao*, Jianhui Wang*, **Xinyuan Song***, Yangfan He, Fangxu Xing, TIANYU SHI, **Free-Mask: A Novel Paradigm of Integration Between the Segmentation Diffusion Model and Image Editing to Improve Segmentation Ability** *ACM Multimedia*, 2025. (*equal contribution)

10. **Xinyuan Song**. **Predicting Stock Price of Construction Companies Using Weighted Ensemble Learning.** *Heylion*, 2023.

11. Tianhao Chen, Xin Xu, Zijing Liu, Pengxiang Li, **Xinyuan Song**, AJAY JAISWAL, Fan Zhang, Jishan Hu, Yang Wang, Hao CHEN, Shizhe Diao, Shiwei Liu, Yu Li, Lu Yin, Can Yang **GPAS: Accelerating Convergence of LLM Pretraining via Gradient-Preserving Activation Scaling** *NeurIPS*, 2025.

12. Colin Zhao, Qinghua Yao, **Xinyuan Song**, Wei Zhu, **ShapLoRA: Allocation of Low-rank Adaption on Large Language Models via Shapley Value Inspired Importance Estimation** *CPAL*, 2026.

13. Yangfan He, Sida Li, Jianhui Wang, **Xinyuan Song**, Kun Li, Xinhang Yuan, Kuan Lu, Menghao Huo, Jingqun Tang, Yi Xin, Jiaqi Chen, Keqin Li, Miao Zhang, Xueqian Wang, **Enhancing Low-Cost Video Editing with Lightweight Adaptors and Temporal-Aware Inversion** *CPAL*, 2026.

14. Guangjing Yang, ZhangYuan Yu, Ziyuan Qin, **Xinyuan Song**, Huahui Yi, Qingbo Kang, Jun Gao, Yiyue Li, Chenlin Du, Qicheng Lao, **Improving Medical Visual Reinforcement Fine-Tuning via Perception and Reasoning Augmentation** *CPAL*, 2026.

15. Yiheng Yao, Zekun Cai, **Xinyuan Song**, Hiroki Hill Kobayashi, Xuan Song, Ryosuke Shibasaki, Liang Zhao **Manifold-Aware Temporal Domain Generalization for Large Language Models** *ICLR CAO Workshop*, 2026.

16. Li Zhang, Mingyu Mei, Ailing Wang, Xianhui Meng, Yan Zhong, **Xinyuan Song**, Liu Liu, Rujing Wang, Zaixing He, Cewu Lu **DICArt: Advancing Category-level Articulated Object Pose Estimation in Discrete State-Spaces** *CVPR*, 2026.

17. Kun Li, Jianhui Wang, Yangfan He, **Xinyuan Song**, Ruoyu Wang, Hongyang He, Wenxin Zhang, Jiaqi Chen, Keqin Li, Sida Li, Miao Zhang, Tianyu Shi, Xueqian Wang, **Optimizing Multi-Round Enhanced Training in Diffusion Models for Improved Preference Understanding** *ICASSP*, 2026.

18. Xuechen Liang, Yangfan He, Yinghui Xia, **Xinyuan Song**, Jianhui Wang, Meiling Tao, Li Sun, Xinhang Yuan, Jiayi Su, Keqin Li, Jiaqi Chen, Jinsong Yang, Siyuan Chen, Tianyu Shi, **Self-evolving Agents with reflective and memory-augmented abilities** *ICASSP*, 2026.

19. Qiang Yi, Yangfan He, Jianhui Wang, **Xinyuan Song**, Shiyao Qian, Xinhang Yuan, Yi Xin, Yijin Wang, Jingqun Tang, Yuchen Li, Junjiang Lin, Hongyang He, Zhen Tian, Tianxiang Xu, Keqin Li, Kuan Lu, Menghao Huo, Jiaqi Chen, Miao Zhang, Tianyu Shi, Jianyuan Ni, **Score: Story coherence and retrieval enhancement for ai narratives**, *LLM&Agents for Recommendation Systems (LARS) Workshop at WWW*, (2026)

20. Yangfan He, Jianhui Wang, Yijin Wang, Yan Zhong, **Xinyuan Song**, Junjiang Lin, Xinhang Yuan, Jingqun Tang, Yi Xin, Hao Zhang, Yuchen Li, Zijian Zhang, Hongyang He, Tianxiang Xu, Miao Zhang, Kuan Lu, Menghao Huo, Keqin Li, Jiaqi Chen, Tianyu Shi, Jianyuan Ni, **Enhancing intent understanding for ambiguous prompt: A human-machine co-adaption strategy** *NeuroComputing*, 2025.

21. Xuechen Liang, Yangfan He, Yinghui Xia, **Xinyuan Song**, Jianhui Wang, Meiling Tao, Li Sun, Xinhang Yuan, Jiayi Su, Keqin Li, Jiaqi Chen, Jinsong Yang, Siyuan Chen, Tianyu Shi, **Self-evolving Agents with Reflective and Memory-Augmented Abilities**, *AAAI 2026 Workshop (LaMAS)*, 2025.

22. Li Zhang, WenBo Xu, Jiankun Wang, **Xinyuan Song**, Qi Wu, Rujing Wang, Xue Wang, Liu Liu., **Towards the Integrated Category-level Articulation Perception and Manipulation Framework for Embodied Intelligence** *IEEE Robotics and Automation Letters (IEEE RA-L)*, 2025.

23. Youhua Li, Yongxin Ni, **Xinyuan Song**, **Bridging NIP and MLM: A Unified Meta-Learning Framework for Personalized Sequential Recommendation** *Transactions on Knowledge Discovery from Data (TKDD)*, 2025. (*equal contribution)

24. Yan Zhong, Xinping Zhao, Li Zhang, **Xinyuan Song**, Tingting Jiang **Adaptive Prompt Learning for Blind Image Quality Assessment with Multi-modal Mixed-datasets Training** *ACM Multimedia*, 2025.

25. Junqiao Wang, Zeng Zhang, Yangfan He, Zihao Zhang, **Xinyuan Song**, Yuyang Song, Tianyu Shi, Yuchen Li, Hengyuan Xu, Kunyu Wu, Xin Yi, Zhongwei Wan, Xinhang Yuan, Zijun Wang, Kuan Lu, Menghao Huo, Jingqun Tang, Guangwu Qian, Keqin Li, Qiuwu Chen, Lewei He, **Enhancing Code LLMs with Reinforcement Learning in Code Generation: A Survey**. *ICLR 2026 Workshop on Logical Reasoning of Large Language Models*, 2026.

26. Dilxat Muhtar*, **Xinyuan Song***, Sebastian Pokutta, Max Zimmer, Nico Pelleriti, Thomas Hofmann, Shiwei Liu, **When Does Sparsity Mitigate the Curse of Depth in LLMs** *ICLR 2026 Workshop DeLTa*, 2026. (*equal contribution)

27. Siyang Zhang, Yecheng Lyu, Luyang Gong, **Xinyuan Song**, Tianyu Shi, **Who Can You Trust on the Road? Adversarial Intentions in Cooperative MARL** *Transportation Research Record*, 2025.

28. Zichen Yuan, Lifan Sun, Yucen Zhuang, Tianqi Xu, Junchen Fu, Hanwen Du, **Xinyuan Song**, Youhua Li, Joemon M. Jose, Yongxin Ni. **AMBER: Adaptive Meta Balanced Paradigm for Heterogeneous Graph-Based Knowledge Tracing** *The Conference on Information and Knowledge Management (CIKM)*, 2025. (*equal contribution)

29. Junbang Liu, Enpei Huang, Dongxing Mao, Hui Zhang, **Xinyuan Song**, Yongxin Ni **ContrastiveGaussian: High-Fidelity 3D Generation with Contrastive Learning and Gaussian Splatting.** *International Conference on Multimedia and Expo (ICME) 2025*, 2025. (*equal contribution)

30. Zichen Yuan, Lifan Sun, Yucen Zhuang, Tianqi Xu, Junchen Fu, Hanwen Du, **Xinyuan Song**, Youhua Li, Joemon M. Jose, Yongxin Ni **SOLAR: Serendipity Optimized Language Model Aligned for Recommendation.** *EMNLP* 2025. (*equal contribution)

31. Yan Zhong, Xingyu Wu, Xinping Zhao, Li Zhang, **Xinyuan Song**, Lei Shi, Bingbing Jiang **Semi-Supervised Multi-Label Feature Selection with Consistent Sparse Graph Learning** *Neural Networks*, 2024.

32. Yan Zhong, Zhiyuan You, Xinping Zhao, Li Zhang, **Xinyuan Song**, Ruoyu Zhao, Lei Shi, Tingting Jiang, **Generalizable Blind Image Quality Assessment with Retrieval Augmentation and Contrastive Vision-Text Local Meta-Training**. *Proceedings of the 34th ACM International Conference on Multimedia (ACM MM)*, 2026.

33. Yundi Hong, Ao Li, **Xinyuan Song**, Yan Zhong, Hongyang He, and Victor Sanchez **Beyond Data Augmentation: Energy-Based Kuramoto Neurons for Semi-Supervised Learning** *CVPR 2026 Workshop DataCV*.

34. Yan Zhong, Zhiyuan You, Xinping Zhao, Li Zhang, **Xinyuan Song**, Ruoyu Zhao, Lei Shi, Tingting Jiang, **Generalizable Blind Image Quality Assessment with Retrieval Augmentation and Contrastive Vision-Text Local Meta-Training** *ACM MM*, 2026.

35. Tianyang Wang, Ming Liu, Benji Peng, **Xinyuan Song**, Charles Zhang, Xintian Sun, Qian Niu, Junyu Liu, Silin Chen, Keyu Chen, Ming Li, Pohsun Feng, Ziqian Bi, Yunze Wang, Yichao Zhang, Cheng Fei, Lawrence KQ Yan **From Bench to Bedside: A Review of Clinical Trials in Drug Discovery and Development** *BIO Integration*, 2025

36. Qian Niu, Junyu Liu, Ziqian Bi, Pohsun Feng, Benji Peng, Keyu Chen, Ming Li, Lawrence K. Q. Yan, Yichao Zhang, Caitlyn Heqi Yin, Cheng Fei, Tianyang Wang, Yunze Wang, Silin Chen, Ming Liu, Ziyuan Qin, Riyang Bao, **Xinyuan Song**, and Zekun Jiang **Large Language Models and Cognitive Science: A Comprehensive Review of Similarities, Differences, and Challenges** *BIO Integration*, 2026.

37. Tianxiang Xu, Junqiao Wang, Yangfan He, **Xinyuan Song**, Peilai Yu, Xunfei Zhu, and Chang Liu **Multi-Scale Volumetric Transformers with Adaptive Uncertainty Modeling for Robust Bacterial Flagellar Motor Localization in Cryo-Electron Tomography** *BIBM*, 2025

Other First Author Papers

1. **Xinyuan Song**, Qingsong Wen, Shirui Pan, Liang Zhao **Complex networks of AI agentic systems: topology, memory, and update dynamics** *Arxiv preprint*, 2026.

2. **Xinyuan Song**, Liang Zhao **Learning to Recommend Multi-Agent Subgraphs from Calling Trees** *Arxiv preprint*, 2026.

3. **Xinyuan Song**, Guangji Bai, Liang Zhao **StructPrune: Structured Global Pruning asymptotics with $\sqrt{N}$ GPU Memory** *Arxiv preprint*, 2026.

4. Siyi Wu, XinLiang Chia, Ziqian Bi, Leyi Zhao, Tianyang Wang, Junhao Song, Yichao Zhang, Keyu Chen, Benji Peng, Xinyuan Song,**AutoSurvey2: Empowering Researchers with Next Level Automated Literature Surveys** *Arxiv preprint*, 2026.

5. (Masterpiece) **Xinyuan Song**, Jiaye Teng, Ziye Ma **Matrix Sensing with Kernel Optimal Loss: Robustness and Optimization Landscape** *Submitted to the Journal of Machine Learning Research (JMLR)*, 2025.

6. Ween Yang*, Jason Liu*, Suli Wang*, **Xinyuan Song***, Lynn Ai, Eric Yang, Tianyu Shi **Lattica: A Decentralized Cross-NAT Communication Framework for Scalable AI Inference and Training** *Arxiv preprint*, 2025. (*equal contribution)

7. Jinji Yang*, **Xinyuan Song***, Ziye Ma, **Recovery Guarantees of the Matrix Lasso Problem: On Equivalence of Convex and Non-Convex Formulations** *Submitted to the NeurIPS*, 2025. (*equal contribution)

8. **Xinyuan Song**, Zeyu Wang, Siyi Wu, Tianyu Shi, Lynn Ai, **Gradientsys: A Multi-Agent LLM Scheduler with ReAct Orchestration** *Submitted to the AAAI*, 2025.

9. **Xinyuan Song**, Yangfan He, Sida Li, Jianhui Wang, Hongyang He, Xinhang Yuan, Ruoyu Wang, Jiaqi Chen, Keqin Li, Kuan Lu, Menghao Huo, Binxu Li, Pei Liu, **Efficient Temporal Consistency in Diffusion-Based Video Editing with Adaptor Modules: A Theoretical Framework** *Submitted to the AAAI*, 2025.

10. Siyi Wu*, Zeyu Wang*, **Xinyuan Song***, Zhengpeng Zhou, Lifan Sun, Tianyu Shi, **GoalfyMax: A Protocol-Driven Multi-Agent System for Intelligent Experience Entities** *Arxiv preprint*, 2025. (*equal contribution)

11. Siyi Wu, Leyi Zhao, Haotian Ma, **Xinyuan Song**, **SDF-TopoNet: A Two-Stage Framework for Tubular Structure Segmentation via SDF Pre-training and Topology-Aware Fine-Tuning** *Submitted to the AAAI* 2025. Corresponding Author

12. Siyi Wu*, Zeyu Wang*, **Xinyuan Song***, Zhengpeng Zhou, Lifan Sun, Tianyu Shi, **GoalfyMax: A Protocol-Driven Multi-Agent System for Intelligent Experience Entities** *Arxiv preprint*, 2025. (*equal contribution)

13. **Xinyuan Song Structured risk model**

14. **Xinyuan Song**, **A design of Convolutional Neural Network model for the Diagnosis of the COVID-19**

Multi-agent system papers

1. **Xinyuan Song**, Zeyu Wang, Siyi Wu, Tianyu Shi, Lynn Ai, **Gradientsys: A Multi-Agent LLM Scheduler with ReAct Orchestration** *Arxiv preprint*, 2025.

2. Ween Yang*, Jason Liu*, Suli Wang*, **Xinyuan Song***, Lynn Ai, Eric Yang, Tianyu Shi **Lattica: A Decentralized Cross-NAT Communication Framework for Scalable AI Inference and Training** *Arxiv preprint*, 2025. (*equal contribution)

3. Siyi Wu*, Zeyu Wang*, **Xinyuan Song***, Zhengpeng Zhou, Lifan Sun, Tianyu Shi, **GoalfyMax: A Protocol-Driven Multi-Agent System for Intelligent Experience Entities** *Arxiv preprint*, 2025. (*equal contribution)

4. Jinwei Su, Yinghui Xia, Qizhen Lan, **Xinyuan Song**, Chen Chen, Yang Jingsong, Lewei He, Tianyu Shi, **Difficulty-Aware Agent Orchestration in LLM-Powered Workflows** *Arxiv preprint*, 2025.

5. Junqiao Wang, Zhaoyang Guan, Guanyu Liu, Tianze Xia, Xianzhi Li, Shuo Yin, **Xinyuan Song**, Chuhan Cheng, Tianyu Shi, Alex Lee, **Meta-Learning Reinforcement Learning for Crypto-Return Prediction** *Arxiv preprint*, 2025.

6. Ji Wang, Kashing Chen, **Xinyuan Song**, Ke Zhang, Lynn Ai, Eric Yang, Bill Shi, **Symphony: A Decentralized Multi-Agent Framework for Scalable Collective Intelligence** *Arxiv preprint*, 2025.

7. Haoran Xu, Jiacong Hu, Ke Zhang, Lei Yu, Yuxin Tang, **Xinyuan Song**, Yiqun Duan, Lynn Ai, Bill Shi, **SEDM: Scalable Self-Evolving Distributed Memory for Agents** *Arxiv preprint*, 2025.

8. Siyi Wu, Junqiao Wang, Zhaoyang Guan, Leyi Zhao, **Xinyuan Song**, Xinyu Ying, Dexu Yu, Jinhao Wang, Hanlin Zhang, Michele Pak, Yangfan He, Yi Xin, Jianhui Wang, Tianyu Shi, **MountainLion: A Multi-Modal LLM-Based Agent System for Interpretable and Adaptive Financial Trading** *Arxiv preprint*, 2025.

Awards

- **Chinese Physics Olympics (CPhO)** silver medal (nation level)
- **Chinese Mathematics Olympics (CMO)** silver medal (nation level)